

Gauge-Equivariant Attention for Lattice Gauge Theory: Learned Variational Operators and Attention Maps as Lattice Observables

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We introduce GELT, a gauge-equivariant lattice transformer whose attention scores are gauge *invariant* and whose value path is matrix bilinear, with keys and values parallel transported over the full L^1 ball of Manhattan radius R along an average of all shortest lattice paths. Exact invariance of the network output is verified to machine precision. Trained on a Rayleigh-quotient loss whose converged value *is* a mass, and restricted to a single timeslice so that the transfer-matrix bound applies, GELT becomes a variational 0^{++} glueball operator for $SU(2)$: on an anisotropic $12^3 \times 24$, $\beta = 2.4$, $\xi = 3$ ensemble it saturates the variational bound at the classical anchor $ma_t \approx 0.33$ and carries significantly more ground-state weight than the optimal combination of a four-level smeared Morningstar–Peardon basis, $\Delta A_0 = +0.078 \pm 0.022$ (3.6σ) combining two independently sampled ensembles, at a consistent mass. Because the attention weights are gauge invariant, any reduction of an attention row is a local scalar lattice operator, so the attention field itself has a mass. In three-dimensional $\mathbb{Z}_2$ gauge theory near criticality the correlation length read out of the attention tracks the classical one at Pearson 0.9966 over a factor 2.7 in ξ and is $12\times$ more sensitive to the evaluation ensemble than to the training coupling. Because 3D $\mathbb{Z}_2$ is *exactly* dual to the 3D Ising model, this can be checked against ground truth rather than against another operator: measured in the dual — where the order parameter, which is the non-local 't Hooft operator on the gauge side and therefore unavailable to any smeared-loop basis, interpolates the same mass gap with near-unit overlap — the trained attention field is the only arm consistent with it ($+3.8\%$) while the classical smeared basis and the untrained network read 9% and 8% low. The structural statement holds already at random initialization; what training adds is ground-state purity, $\Delta A_0 = +0.11$ to $+0.27$ (5.4σ – 23.5σ), and it is that purity the dual confirms as accuracy. Equivariant attention maps are therefore not diagnostics but measurements.

I. INTRODUCTION

Lattice gauge theory turns quantum field theory into a statistical problem on a finite grid [1], and the quantities it must estimate — masses, string tensions, topological susceptibilities — are all expectation values of gauge-invariant operators. Two facts make machine learning attractive here and one makes it delicate. The first fact is that the accuracy of a lattice measurement is usually limited not by statistics but by *operator quality*: how much of a chosen interpolating operator's spectral weight sits on the state one wants. The second is that the search for a good operator — choose loop shapes, choose a smearing schedule, diagonalize a small correlator matrix — is an optimization problem with a differentiable objective. The delicate part is symmetry: a network that is not gauge equivariant computes a function of the gauge in which the configuration happens to be written, and no gauge-fixing convention makes such a function an observable.

Lattice gauge equivariant convolutional networks (L-CNNs) resolved the symmetry problem [17], by building the network out of parallel transports and bilinear loop products so that equivariance holds exactly and by construction; related gauge-covariant architectures followed [18, 19]. Their convolutional kernels, however, are fixed after training: the same stencil is applied at every site of every configuration. Attention [20] removes

exactly that restriction — the aggregation weights are computed from the content in front of the network — but a naive attention score over gauge fields is not an observable.

This paper develops GELT (gauge-equivariant lattice transformer), which resolves the tension with two design choices. The attention score is a gauge-*invariant* two-loop trace between a query and a parallel-transported key, and the value path is matrix *bilinear*, $\propto Q_v^\dagger \tilde{V}$, so that the L-CNN loop-doubling universality argument transfers directly and each layer roughly doubles the maximum loop length the network can express. Transport between sites is averaged over *all* shortest lattice paths in the L^1 ball of radius R , computed by a dynamic-programming recursion rather than by enumeration, so a single block already sees a rotationally symmetric, non-axis-aligned receptive field.

Two results follow, and they are of different kinds.

As an operator, GELT is trained without labels on the Rayleigh quotient $C(1)/C(0)$ of the transfer matrix, whose converged value is the mass itself. Constrained to act on a single timeslice — the same condition that forbids temporal smearing, applied to the network — it becomes a legitimate interpolating operator at every step of training, and on an anisotropic $SU(2)$ ensemble it beats a four-level smeared Morningstar–Peardon basis [15] on ground-state overlap while agreeing with it on the mass, as the variational principle demands.

As a *measurement*, the attention map is an observable in its own right. Gauge invariance of the score means any per-site reduction of an attention row is a local gauge-invariant scalar field; single-timeslice locality means its zero-momentum correlator has an exact spectral decomposition. The attention field therefore has a mass, and that mass is the theory’s mass gap. We verify this in three-dimensional $\mathbb{Z}_2$ gauge theory, where the correlation length can be tuned over a real dynamic range, and we are careful to separate what this establishes about the architecture from what it establishes about learning.

II. THE GELT ARCHITECTURE

A. Fields and symmetry

Gauge fields live on links, $U_\mu(x) \in G$, and a gauge transformation is a group element $\Omega(x)$ at every site,

$$U_\mu(x) \rightarrow \Omega(x) U_\mu(x) \Omega^\dagger(x + \hat{\mu}). \quad (1)$$

The network’s input is the untraced plaquette field

$$P_{\mu\nu}(x) = U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x), \quad (2)$$

one channel per plane $\mu < \nu$, which transforms in the *adjoint* way, $W(x) \rightarrow \Omega(x) W(x) \Omega^\dagger(x)$ — the hidden state of every layer keeps this transformation law. Color indices are carried explicitly at every stage, even for $\mathbb{Z}_2$ where $n_c = 1$, so that every matrix product and every dagger is written out and the implementation ports verbatim from $\mathbb{Z}_2$ to $SU(N)$.

B. Shortest-path-averaged transport

To compare fields at x and $x + \Delta x$ one must transport one to the other. GELT uses, for every signed offset in the L^1 ball $1 \leq |\Delta x|_1 \leq R$, the average over *all* shortest lattice paths,

$$T_{\Delta x}(x) = \frac{1}{N_{\Delta x}} \sum_{\substack{\mathcal{P}: x \rightarrow x + \Delta x \\ |\mathcal{P}| = |\Delta x|_1}} U_{\mathcal{P}}, \quad N_{\Delta x} = \frac{|\Delta x|_1!}{\prod_\mu |\Delta x_\mu|!}, \quad (3)$$

built by the recursion

$$\begin{aligned} T_{\Delta x}(x) = & \sum_{\mu: \Delta x_\mu > 0} U_\mu(x) T_{\Delta x - \hat{\mu}}(x + \hat{\mu}) \\ & + \sum_{\mu: \Delta x_\mu < 0} U_\mu^\dagger(x - \hat{\mu}) T_{\Delta x + \hat{\mu}}(x - \hat{\mu}), \end{aligned} \quad (4)$$

evaluated in one pass ordered by $|\Delta x|_1$, so the cost is linear in the number of offsets rather than in the (exponentially many) paths. Under (1) every term transforms identically, $T_{\Delta x}(x) \rightarrow \Omega(x) T_{\Delta x}(x) \Omega^\dagger(x + \Delta x)$, so the average is gauge covariant; unlike a single axis-aligned path

it also preserves the lattice rotation symmetry. The offset ball is the network’s receptive field per layer: $|\Delta x|_1 \leq R$ contains 24 offsets in 3D at $R = 2$ and 84 offsets in 2D at $R = 6$.

C. The equivariant attention block

One GELT block (GEMHSA, Fig. 1) maps a covariant field W to a covariant field of the same shape:

(i) *Augment and project.* The channel axis is extended by the on-site identity and the daggers, $C \rightarrow 2C + 1$, which lets a single linear map express both a field and its inverse; four per-head projections Q_s, Q_v, K, V of width d_{qkv} are then taken with complex weights shared over sites. All four remain adjoint-covariant.

(ii) *Transport.* Keys and values at $x + \Delta x$ are brought to x , $\tilde{K} = T_{\Delta x} K T_{\Delta x}^\dagger$, so that they conjugate with $\Omega(x)$, like the query.

(iii) *Invariant score.* The score is a real trace,

$$s_{x \rightarrow x + \Delta x} = \frac{1}{\sqrt{n_c d_{qkv}}} \operatorname{Re} \sum_a \operatorname{Tr} [Q_{s,a}^\dagger(x) \tilde{K}_a(x + \Delta x)], \quad (5)$$

which is *exactly gauge invariant*, the two $\Omega(x)$ factors cancelling under the trace. Equation (5) is the matrix generalization of the usual inner product $q^\dagger k$, and it is a familiar lattice object: a two-loop correlator of the kind that measures glueball propagators and string tensions. Relative position enters through a rotary encoding [21] acting on channel pairs of K with an offset-dependent angle, which makes the score offset selective without breaking invariance (it acts in channel space, not color space).

(iv) *Softmax and bilinear value path.* With $\alpha = \operatorname{softmax}_{\Delta x}(s)$,

$$W^{\text{out}}(x) = \sum_{\Delta x} \alpha_{x \rightarrow x + \Delta x} Q_v^\dagger(x) \tilde{V}(x + \Delta x), \quad (6)$$

a product of two covariant matrices, hence covariant. This is the key departure from standard attention: because the value path multiplies transported loop matrices, every layer composes longer Wilson loops, and loop degree doubles with depth — the mechanism behind L-CNN’s universality argument [17].

(v) *Mix, gate, residual.* A complex linear map returns the head outputs to C channels; the sublayer output is gated by a trace-activated real scalar, $g = \operatorname{softplus}(\operatorname{Re} \operatorname{Tr} W_{\text{mix}}/n_c)$, and added to the untouched residual stream. A trace and a per-site MLP turn the final field into a gauge-invariant scalar per site, i.e. exactly an operator density; a spatial reduction is optional.

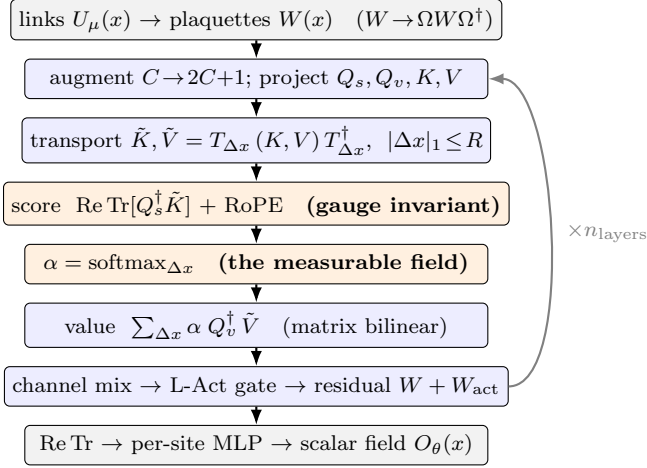


FIG. 1. One GELT block. Blue stages carry adjoint-covariant matrix fields; orange stages are gauge singlets. The scores, and hence the attention weights α , are exactly gauge invariant — which is what makes them observables (Sec. VI) rather than diagnostics. The value path multiplies two covariant matrices, so loop degree doubles with depth.

D. Exactness of the symmetry

Equivariance is not approximate and not learned. On random $SU(2)$ configurations with a Haar-random gauge transformation $\Omega(x)$ generated by QR decomposition, the full network output satisfies $O_\theta[U^\Omega] = O_\theta[U]$ with a maximum drift of 0.0 — bit-identical outputs — for both positional-encoding variants in **complex128** and in **complex64**, at 4 layers and nonzero readout. A unit-test suite of 111 tests covers the block equivariance $\text{block}(W^\Omega, T^\Omega) = \Omega \text{block}(W, T) \Omega^\dagger$ for $SU(2)$, $SU(3)$ and $\mathbb{Z}_2$, the covariance of the transport (3), and the octant identity $T_{-\Delta x}(x) = T_{\Delta x}^\dagger(x - \Delta x)$.

E. Cost

Memory, not arithmetic, is the binding constraint: the transport tensor is $(N_{\text{cfg}}, n_{\text{off}}, |\Lambda|, n_c, n_c)$ and n_{off} grows like R^D . Two mitigations are used throughout. Restricting the network to a single timeslice (Sec. VC), which the physics demands anyway, collapses the 4D transport to a 3D one and cuts the cost by a factor ≈ 70 ; and transports are rebuilt per batch on the fly rather than cached, which is cheap — the recursion (3) builds the whole $R = 2$ ball for 2000 8^2 configurations in ≈ 0.02 s — so memory is traded for a negligible constant.

III. ENSEMBLES AND SAMPLING

A. Actions

All ensembles use the Wilson action $S = \beta \sum_{x, \mu < \nu} (1 - \frac{1}{n_c} \text{Re Tr } P_{\mu\nu})$. Spectroscopy additionally needs an *anisotropic* lattice, $a_t = a_s/\xi$ with $\xi > 1$, so that a heavy state’s exponential decay is sampled by ξ times more timeslices [15]. At tree level this is a reweighting by plaquette orientation,

$$\beta_s = \beta/\xi \text{ (spatial)}, \quad \beta_t = \beta \xi \text{ (temporal)}, \quad (7)$$

which we fold into the staple sum so that $\xi = 1$ reproduces the isotropic code path bit for bit.

B. Exact local updates

$SU(2)$ admits two exact, tuning-free local updates whose combination beats Metropolis critical slowing down. The heat bath [7, 8] exploits the fact that a sum of $SU(2)$ elements is proportional to an $SU(2)$ element, $A = kV$, so the single-link conditional distribution is known in closed form and the new link is independent of the old. Overrelaxation [9] reflects $U \rightarrow V^\dagger U^\dagger V^\dagger$, preserving the action exactly while moving as far as possible across the constant-action surface; since the reflection is expansive off the group manifold, each link is re-projected onto $SU(2)$ in closed form. The production sweep is one heat-bath sweep followed by $n_{\text{OR}} = 4$ overrelaxation sweeps in a checkerboard pattern.

For $\mathbb{Z}_2$ the analogous exact update is $P(U = +1) = \sigma(2\beta s)$ with s the staple sum. It is not a convenience but a necessity near criticality: the single-link Metropolis proposal is the flip $U \rightarrow -U$, whose acceptance collapses to ≈ 0.02 as the configuration orders, freezing the chain exactly where the correlation length is largest.

C. Validation and autocorrelation

Figure 2 shows the standard checks for $SU(2)$: thermalization, the exactly known two-dimensional mean plaquette $I_2(\beta)/I_1(\beta)$ reproduced across three volumes, the three-dimensional confining crossover, and the plaquette autocorrelation ($\tau_{\text{int}} \approx 4.5$ sweeps for Metropolis at $\beta = 0.8$). The anisotropic action is validated separately (Fig. 3): at $\xi = 1$ it returns the exact two-dimensional plaquette, 0.4329 ± 0.0017 against 0.4331 (0.1σ); at $\xi > 1$ the plaquette splits in the intended direction, $\langle P_{st} \rangle$ rising and $\langle P_{ss} \rangle$ falling; and a Creutz-ratio estimate gives a *renormalized* anisotropy $\xi_R \approx 3.32$ at bare $\xi = 3$ — the expected tree-level mismatch, measured rather than tuned away.

TABLE I. Ensembles. All errors quoted in this paper are jackknife estimates over configurations, blocked where indicated.

theory	lattice	coupling	N	use
$SU(2)$, 4D	$12^3 \times 24$	$\beta=2.4, \xi=3$	2000	anchor + training
$SU(2)$, 4D	$12^3 \times 24$	$\beta=2.4, \xi=3$	2000	replication
$SU(2)$, 4D	$12^3 \times 24$	$\beta=2.4, \xi=3$	32	attention readout
$SU(2)$, 4D	$12^3 \times 24$	$\beta \in [2.1, 2.7]$	2000	ξ_s scan
$\mathbb{Z}_2$, 3D	$24^2 \times 48$	$\beta \in [0.745, 0.76]$	2000	attention operator
$SU(2)/\mathbb{Z}_2$, 2D	8^2	$\beta=2.4/0.7$	2000	regression

$SU(2)$: heat bath + 4 overrelaxation sweeps, $n_{\text{skip}} = 5$. $\mathbb{Z}_2$: exact heat bath, $n_{\text{skip}} = 200$. 2D regression sets: Metropolis.

Configurations are separated by n_{skip} sweeps set from the integrated autocorrelation time of the *measured* observable (Madras–Sokal windowing [16]), not of the plaquette: extended operators decorrelate more slowly. For the anisotropic $SU(2)$ ensembles the smeared glueball operator gives $\tau_{\text{int}} \approx 2$ and we take $n_{\text{skip}} = 5$; for $\mathbb{Z}_2$ near β_c critical slowing down ($\tau \sim \xi^z$, $z \approx 2$) gives $\tau_{\text{int}} \approx 44$ sweeps and we take $n_{\text{skip}} = 200$. Errors are always jackknife, blocked when residual autocorrelation is suspected, and comparisons between two operators measured on the same configurations are always jackknifed as a *difference*, so that the shared ensemble fluctuations cancel.

Table I lists every ensemble used below.

IV. SUPERVISED BENCHMARKS

Before using GELT as an operator we check what it can represent, on targets whose answer is known exactly.

The inductive-bias gap. The motivating observation is that a non-equivariant convolutional network fed *links* cannot learn the Wilson action ($R^2 \approx 0$ across volumes on Haar-random $\mathbb{Z}_2$ data), because the target requires multiplying four specific link matrices around a plaquette — a product an additive kernel does not express — while the same network fed *plaquettes* reaches $R^2 \approx 0.99$, the task having collapsed to a linear sum. The gap is precisely what an equivariant transport supplies. (On Haar data even a perfect regressor only memorizes the action *function*; the β -dependent physics below requires the Monte Carlo data path of Sec. III.)

An exactly representable target. The naive topological charge density $q(x)$ is a single matrix bilinear in the plaquettes at x . GELT’s value path (6) is matrix bilinear, so the target lies inside the hypothesis class algebraically. A 10 673-parameter model (20 241 real degrees of freedom) at $L = 8$, $D = 4$, $\beta = 2.4$, $R = 2$ reaches $R^2 = 1.0000$ on held-out $SU(2)$ configurations, with the training loss reaching the float32 floor. This is a representation check rather than a learning result, and its lesson is stated as such: a target that needs no context

TABLE II. Test R^2 for per-site $R \times T$ Wilson-loop regression at fixed receptive radius $R = 2$ on 8^2 lattices, $N = 2000$ configurations, 40 epochs. “avg” is the shortest-path average of Eq. (3), “single” one canonical shortest path, “proj” the average projected back onto the group. Accuracy is governed by the receptive radius, not by the transport mode.

loop	$\mathbb{Z}_2, \beta = 0.7$		$SU(2), \beta = 2.4$		
	avg	single	avg	single	proj
1×1	0.99999	0.99999	1.000	1.000	1.000
2×2	0.871	0.799	0.979	0.988	0.972
3×3	0.299	0.259	0.022	−0.005	−0.004

is fit with no context.

Receptive field and transport mode. Rectangular $R \times T$ Wilson loops probe the opposite regime, where the target’s support exceeds one block’s reach. Table II scans loop size against the two transport modes of Eq. (3) at fixed $R = 2$, in both gauge groups. The pattern is unambiguous and is about geometry, not about paths: the 1×1 loop is fit exactly, the 2×2 loop — whose extent already matches the ball — is fit well, and the 3×3 loop, which no depth-limited $R = 2$ stack can close, is not fit at all. Between the shortest-path average and a single canonical path there is no consistent difference: the average is ahead by 0.07 and 0.04 in R^2 on $\mathbb{Z}_2$ and behind by 0.01 on $SU(2)$, i.e. within run-to-run scatter. Path averaging is therefore adopted for its symmetry — it preserves lattice rotation invariance, which a canonical path breaks — and not for accuracy; projecting the averaged transport back onto the group, a third variant, costs a transport build of 2.9 s and buys nothing.

V. A NETWORK AS A VARIATIONAL OPERATOR

A. Masses, and why glueballs are hard

With periodic temporal extent N_t and transfer matrix $\hat{T} = e^{-a_t \hat{H}}$ [5], the connected correlator of a zero-momentum operator $\bar{O}(t) = \sum_{\vec{x}} O(\vec{x}, t)$ decomposes as

$$C(\Delta) = \langle \delta \bar{O}(t+\Delta) \delta \bar{O}(t) \rangle = \sum_{n>0} |\langle 0 | \hat{O} | n \rangle|^2 e^{-m_n \Delta a_t} + \dots, \quad (8)$$

the ellipsis being the periodic image. Every term is positive, so $m_{\text{eff}}(\Delta) = \log[C(\Delta)/C(\Delta+1)]$ approaches the ground-state mass *from above*: the correlator is a variational bound machine. The 0^{++} glueball is the hard case [6]: its variance is itself a vacuum-channel correlator, so noise is Δ -independent while signal decays, giving $S/N \sim \sqrt{N} e^{-m_G \Delta a_t}$ [11]. Since $m_G a$ is of order one, the signal is gone within a few slices and more statistics barely help. The plateau must instead be dragged to

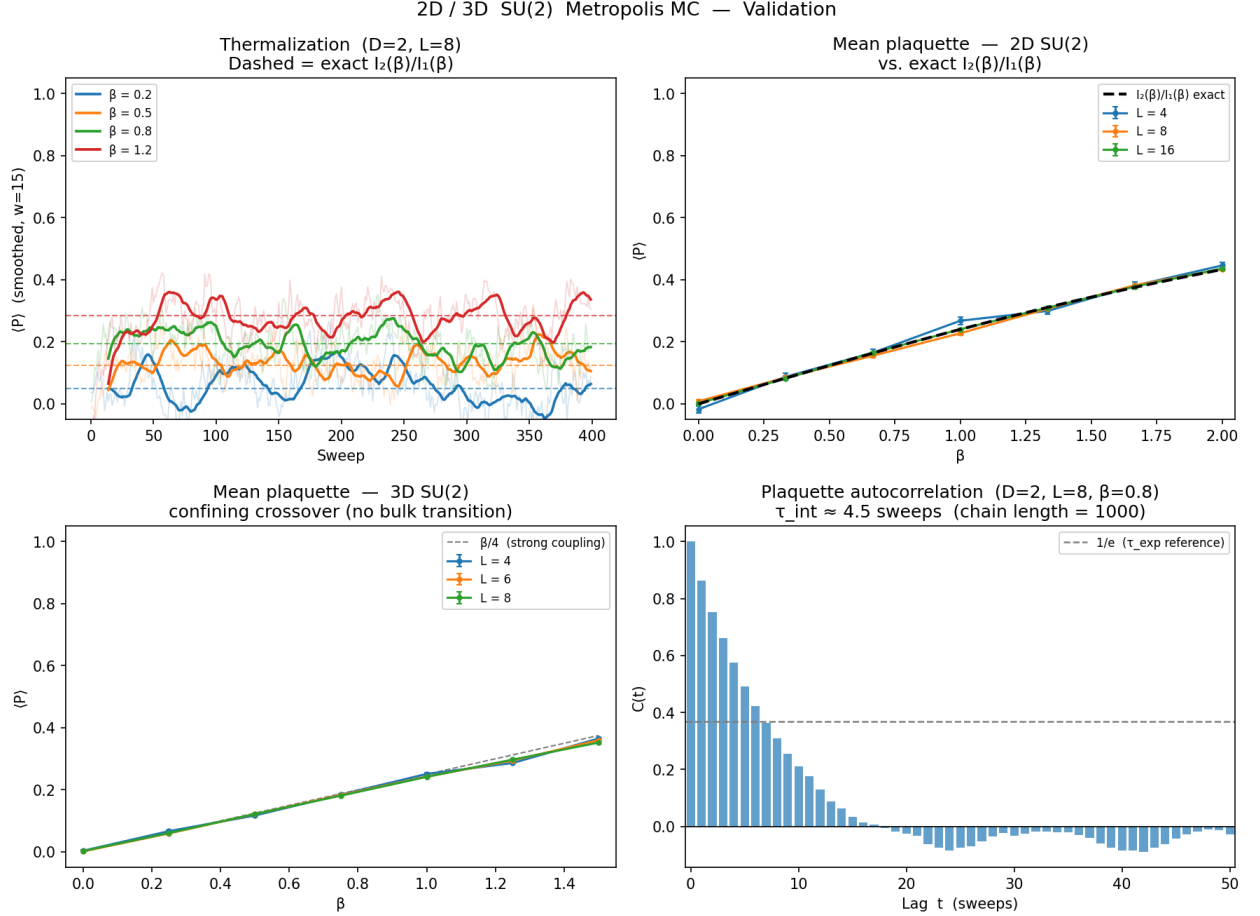


FIG. 2. $SU(2)$ Monte Carlo validation. Top left: thermalization of the mean plaquette at four couplings against the exact $I_2(\beta)/I_1(\beta)$ (dashed). Top right: the two-dimensional mean plaquette reproduces the exact curve at $L = 4, 8, 16$. Bottom left: the three-dimensional confining crossover, with the strong-coupling $\beta/4$ asymptote. Bottom right: plaquette autocorrelation, $\tau_{\text{int}} \approx 4.5$ sweeps. The equivalent four-panel check for $\mathbb{Z}_2$ reproduces $\langle P \rangle = \tanh \beta$ in two dimensions and the transition near $\beta_c \approx 0.761$ in three.

small Δ , which means building an operator whose spectral weight already sits on the ground state — the entire craft of glueball spectroscopy, classically executed with APE smearing [10] and a generalized eigenvalue problem (GEVP) [12–15] over a basis of smearing levels.

B. The Rayleigh loss

Let O_θ be any gauge-invariant scalar field built from the spatial links of timeslice t , and $|\psi\rangle = (\hat{O}_\theta - \langle \hat{O}_\theta \rangle)|0\rangle$. Then $C(\Delta) = \langle \psi | \hat{T}^\Delta | \psi \rangle$ with $\langle \psi | 0 \rangle = 0$, so

$$R(\theta) = \frac{C(1)}{C(0)} = \frac{\langle \psi | \hat{T} | \psi \rangle}{\langle \psi | \psi \rangle} \leq e^{-m_G a t}, \quad (9)$$

a Rayleigh quotient of the transfer matrix restricted to the non-vacuum sector. Training on $\mathcal{L} = -C(1)/C(0)$ is therefore variational: no labels, no supervision, the training signal is the same Monte Carlo correlator the

classical analysis measures, the bound holds at every step because equivariance makes O_θ a legitimate operator for *any* θ , and at convergence $m_G a t = -\log R$ — the converged loss *is* the mass. In practice we average two ratios, $\mathcal{L} = -\frac{1}{2}[C(1)/C(0) + C(2)/C(0)]$, whose floor at $ma_t = 0.33$ is ≈ -0.62 .

C. The single-timeslice constraint

Equation (9) required $\bar{O}(t)$ to depend on timeslice t alone — the same condition that forbids smearing in time. A four-dimensional network violates it twice, through temporal plaquettes in its input and through a receptive field that leaks across slices, and the violation is not a small correction: it makes the loss *gameable*. If $O(t)$ may depend on neighboring slices, the cheapest route to $C(1)/C(0) \rightarrow 1$ is an output that varies slowly in t ; in the limit a per-configuration constant gives a fake mass

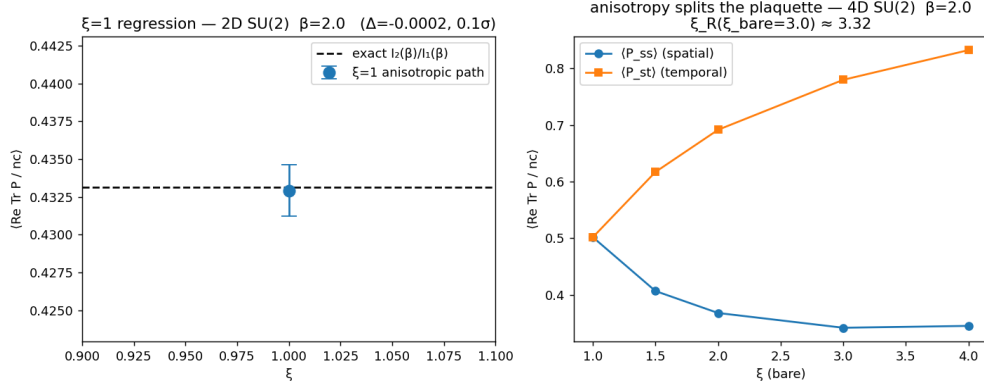


FIG. 3. The anisotropic action. Left: at $\xi = 1$ the anisotropic code path reproduces the exact two-dimensional $SU(2)$ plaquette to 0.1σ . Right: at $\xi > 1$ temporal plaquettes grow and spatial ones shrink, as Eq. (7) requires; a Creutz-ratio estimate returns $\xi_R \approx 3.32$ at bare $\xi = 3$.

of zero. We therefore run GELT as a *three-dimensional* network on the spatial links of one timeslice, batched over (configuration $\times$ timeslice). Within its slice the operator keeps its full spatial loop-building power, which is exactly the domain the classical operator uses. The constraint that protects the physics also pays the memory bill (Sec. II).

D. Making the loss trainable

Three pathologies had to be removed, each of which is a property of the objective rather than of the architecture. (i) *Zero gradient at initialization*: $C(0)$ and $C(1)$ are both quadratic in the output, so a zero-initialized readout — a common stabilization — sits at an exact saddle and training never starts; the readout must be initialized nonzero. (ii) *Denominators*: chaining ratios $C(\Delta)/C(\Delta-1)$ places the small, signed, noisy $C(1)$ in a denominator, which keeps values finite but makes gradients meaningless; every ratio uses the large positive $C(0)$, and the correlator arithmetic runs in float64. (iii) *The scale-flat direction*: the quotient is invariant under $O \rightarrow \lambda O$, an exactly flat direction along which the bilinear value path's squared per-layer gain drives $C(0)$ to float32 overflow once the inputs are smooth. Adding $10^{-2}(\log C(0))^2$ pins the scale without touching the ratio, hence without biasing the physics. Finally, the bound holds in expectation and not on a finite sample, so all reported numbers come from a hard split (70/10/20, contiguous in Monte Carlo time) with checkpoint selection on validation and blocked jackknife on the untouched test configurations.

E. The classical anchor

The classical measurement (Fig. 4) is the orientation-summed spatial Wilson loop at APE levels 0, 2, 4, 6, analyzed with the connected correlator, m_{eff} , and a GEVP at $t_0 = 1$. On the isotropic 12^4 lattice there is no plateau at all — the signal dies by $\Delta \approx 3$ and no operator basis can repair a lattice — while on the anisotropic $12^3 \times 24$, $\xi = 3$ ensemble $C(\Delta)$ decays cleanly over 10–12 slices and the GEVP ground state gives

$$m_{\text{eff}}(1) = 0.365 \pm 0.008, \quad m_{\text{eff}}(2) = 0.333 \pm 0.011, \quad (10)$$

confirmed at $m_{\text{eff}}(3) = 0.333 \pm 0.036$ on the held-out split, i.e. an anchor $ma_t \approx 0.33$. We stress what this number is not: with $\beta_s = \beta/\xi = 0.8$ the spatial lattice is coarse, so $ma_s = \xi ma_t \approx 1.0$ is not continuum physics. The claim established is that a plateau exists on this ensemble, which is what the learned operator must be judged against.

F. Result: a learned operator that wins on overlap

Fed *thin* plaquettes, the learned operator is genuine but loses: its m_{eff} descends $0.705 \rightarrow 0.527 \rightarrow 0.441 \rightarrow 0.310$ and it far outperforms the thin classical operator, but it sits above the smeared GEVP at small Δ and, appended to the classical basis as a fifth operator, changes the GEVP by fractions of an error bar. The diagnosis is a depth budget — four bilinear layers compose loops of degree ≤ 16 , while six APE steps contain branched staple content of degree $\sim 3^6$ within the same radius — and the correlation of the learned operator with the smearing ladder peaks at APE $\times 2$ ($|r| = 0.68, \mathbf{0.87}, 0.71, 0.59$ for levels 0, 2, 4, 6): the network relearned about two smearing steps.

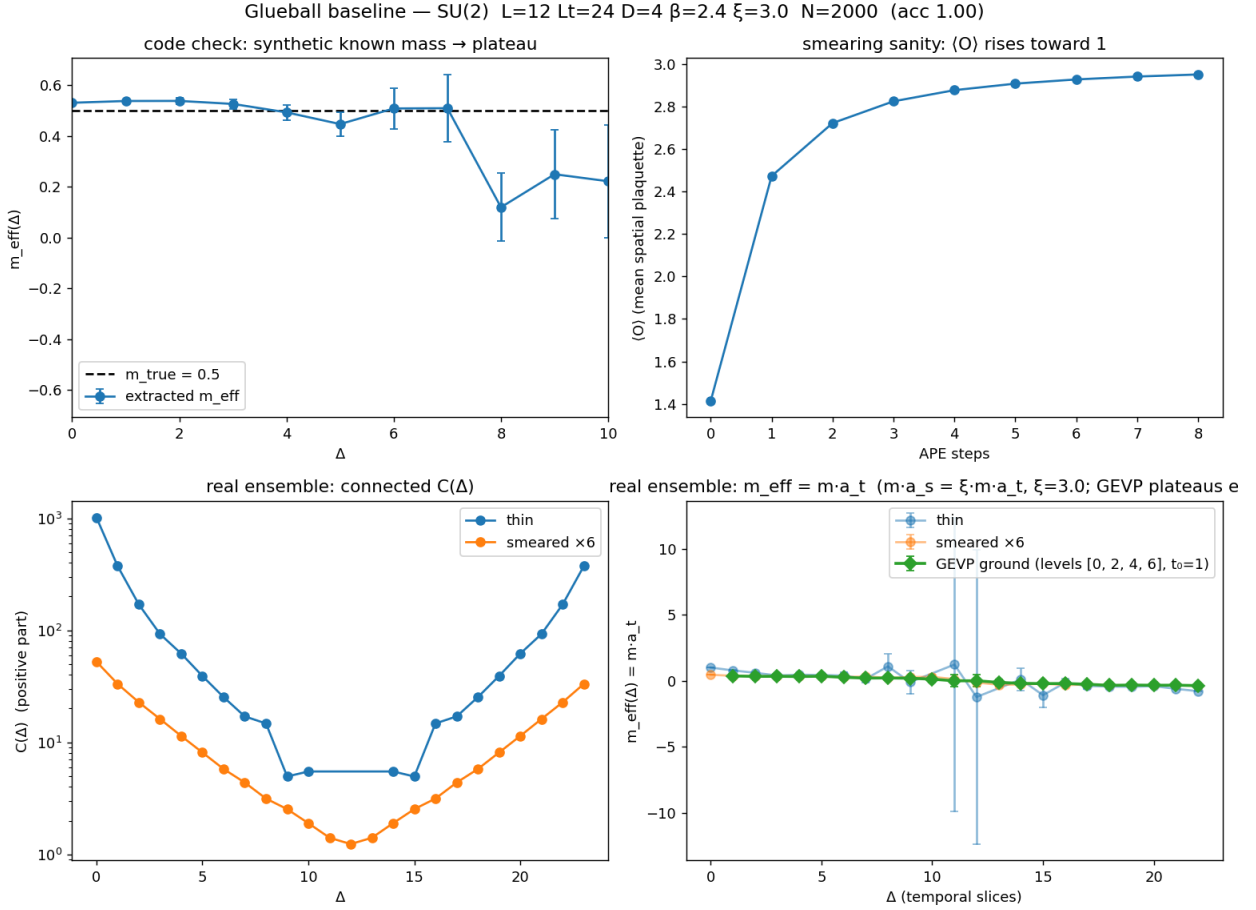


FIG. 4. Classical 0^{++} baseline on the anisotropic ensemble ($12^3 \times 24$, $\beta = 2.4$, $\xi = 3$, $N = 2000$). Top left: the analysis chain recovers a known mass from a synthetic single-exponential correlator. Top right: smearing monotonicity. Bottom left: the connected correlator decays over 10–12 slices, symmetric about $\Delta = N_t/2$. Bottom right: effective masses; the multi-level GEVP ground state (green) plateaus near 0.33 where the single operators do not, and stays resolved where the thin operator’s errors explode.

The fix is to supply the smearing and let the network do what smearing cannot. Feeding plaquettes built at the same cumulative APE levels the GEVP uses (0, 2, 4, 6; 3 planes $\times$ 4 levels = 12 input channels), with transport still built from thin links, makes the comparison maximally sharp: both methods see *exactly* the same smearing levels, the GEVP may take one global linear combination of them, and the network may mix them nonlinearly and site by site. Any gap is then attributable to the network. Training is stable with the scale pin and reaches a best validation loss of -0.6185 , i.e.

$$ma_t \approx 0.329 : \quad (11)$$

the Rayleigh quotient *saturates the transfer-matrix bound at the anchor mass*. Table III and Fig. 5 give the test-split comparison.

Because both operators are measured on the same configurations their errors are correlated, and the honest test

TABLE III. Test effective masses in units of a_t , blocked jackknife on the 400 held-out configurations. The learned operator plateaus from $\Delta = 1$; the enlarged five-operator GEVP collapses onto it.

operator	$m_{\text{eff}}(1)$	$m_{\text{eff}}(2)$	$m_{\text{eff}}(3)$	$m_{\text{eff}}(4)$
GELT (learned)	0.370(16)	0.333(20)	0.345(29)	0.336(50)
classical GEVP	0.398(16)	0.357(25)	0.333(36)	0.284(50)
GEVP + GELT	0.366(17)	0.333(20)	0.347(30)	0.319(48)

is a blocked jackknife of the *difference*:

$$\Delta m_{\text{eff}}|_{\Delta=1} = -0.028 \pm 0.007 \text{ (3.9}\sigma\text{)}, \quad (12)$$

with -0.024 ± 0.014 at $\Delta = 2$ and $+0.012 \pm 0.021$ at $\Delta = 3$. The learned operator sits significantly below the classical GEVP exactly where excited-state contamination lives, and is statistically identical on the plateau — a better operator measuring the same physics, which is

what the variational principle demands of a legitimate win. Two corroborating facts: offered the four-level ladder *plus* GELT, the variational solver picks essentially pure GELT (the enlarged basis matches GELT’s own m_{eff} to ~ 0.004 at $\Delta = 1$); and the ladder-independent part of the learned operator’s variance *grew* from $\sim 8\%$ to $\sim 18\%$ during training while the loss improved, so the win is driven by the network’s own content and not by residual noise.

G. Quoting the result: cosh fits and the overlap A_0

Effective masses are how lattice results are *plotted*; masses are *quoted* from a fit, and operator quality — what this comparison is actually about — is quoted as the ground-state overlap, the number Morningstar and Peardon use to rank smeared operators [15]. We fit each connected correlator to the periodic single-state form

$$C(\Delta) \approx A[e^{-ma_t\Delta} + e^{-ma_t(N_t-\Delta)}] \quad (13)$$

over one *shared* window $\Delta \in [2, 7]$ — comparability beats per-operator optimality, and starting at $\Delta = 2$ keeps the GEVP’s residual $\Delta = 1$ contamination out of its own mass fit, since that contamination is what the overlap should report. Extrapolating the fitted ground state back to $\Delta = 0$ gives

$$A_0 = \frac{A(1 + e^{-ma_t N_t})}{C(0)} \in (0, 1], \quad (14)$$

the fraction of the operator’s spectral weight carried by the glueball. This closes a conceptual loop: the Rayleigh quotient is, up to the wrap term, a weighted average $\sum_n w_n e^{-m_n a_t}$ with $w_{\text{ground}} = A_0$, so saturating the variational bound *is* the statement $A_0 \rightarrow 1$. Objective and metric coincide.

The right comparator for a single learned operator is not a whole basis but the best single operator that basis contains, which the GEVP supplies: the ground-state eigenvector v_0 at $(t_0, t_d) = (1, 2)$ defines the fixed-vector *projected* operator $\sum_i (v_0)_i \bar{O}_i$ [14]. The entire fit, v_0 included, is redone inside each delete-block jackknife sample. Table IV and Fig. 6 give the result, with the correlated differences

$$\Delta(ma_t) = -0.008 \pm 0.014, \quad \Delta A_0 = +0.066 \pm 0.031 \quad (2.1\sigma). \quad (15)$$

In the field’s own language: *both operators measure the same mass, but the learned operator carries 90% of its spectral weight on the ground state against 84% for the optimal combination of the entire classical basis.* Note also that the GEVP gains only ~ 0.03 of overlap over its best single member, while one learned operator gains ~ 0.10 over the whole basis.

TABLE IV. Cosh fits (13) on the shared window $\Delta \in [2, 7]$, blocked jackknife on 400 held-out configurations, for the primary ensemble and for an independently sampled replication ensemble. χ^2/dof is artificially low because neighboring $C(\Delta)$ are strongly correlated while the χ^2 is diagonal; it is a fit-quality heuristic, and the quoted errors come from the jackknife, which propagates the correlations correctly.

ensemble	operator	ma_t	A_0	χ^2/dof
primary	GELT (learned)	0.332(27)	0.903(47)	0.05
	GEVP-projected	0.340(30)	0.837(56)	0.04
	APE $\times 6$	0.340(32)	0.805(52)	0.11
replication	GELT (learned)	0.374(31)	1.013(62)	0.01
	GEVP-projected	0.400(38)	0.925(71)	0.08
	APE $\times 6$	0.390(37)	0.901(68)	0.05

H. Replication on an independent ensemble

The entire chain — sampling, from-scratch training, checkpoint selection, test-split analysis — was repeated with a different sampler seed, nothing tuned on the new data. The operator-quality result reproduces (Table IV, Fig. 7): $\Delta A_0 = +0.089 \pm 0.030$ (2.9σ), with a correlated m_{eff} difference of -0.038 ± 0.008 (4.5σ) at $\Delta = 1$ — the same pattern, if anything stronger. The two independent ensembles combine to

$$\Delta A_0^{\text{comb}} = +0.078 \pm 0.022 \quad (3.6\sigma). \quad (16)$$

What does *not* replicate is equally instructive. On the new ensemble every operator, classical and learned alike, reads $\sim 1\sigma$ heavier, and training converged to a best Rayleigh loss of -0.564 , decoding to $m \approx 0.395$ — saturated to ~ 0.004 , exactly as before, but of a different floor. The Rayleigh floor is the variational bound *as realized on the sampled data*, an empirical property of the ensemble: loss values do not replicate between independent ensembles, operator-quality statements do.

VI. THE ATTENTION MAP AS A LATTICE OPERATOR

A. The claim

The attention weights $\alpha_{x \rightarrow x+\Delta x}$ are computed from the invariant score (5), so they are gauge invariant at every site. Therefore *any* reduction of an attention row to a scalar,

$$A(x) = f(\alpha_{x \rightarrow \cdot}), \quad (17)$$

is a local gauge-invariant scalar field built from the links — an ordinary lattice operator. We use three qualitatively different reductions: the self weight $\alpha_{x \rightarrow x}$, the radial first moment $\sum_{\Delta x} |\Delta x|_1 \alpha$, and the offset entropy

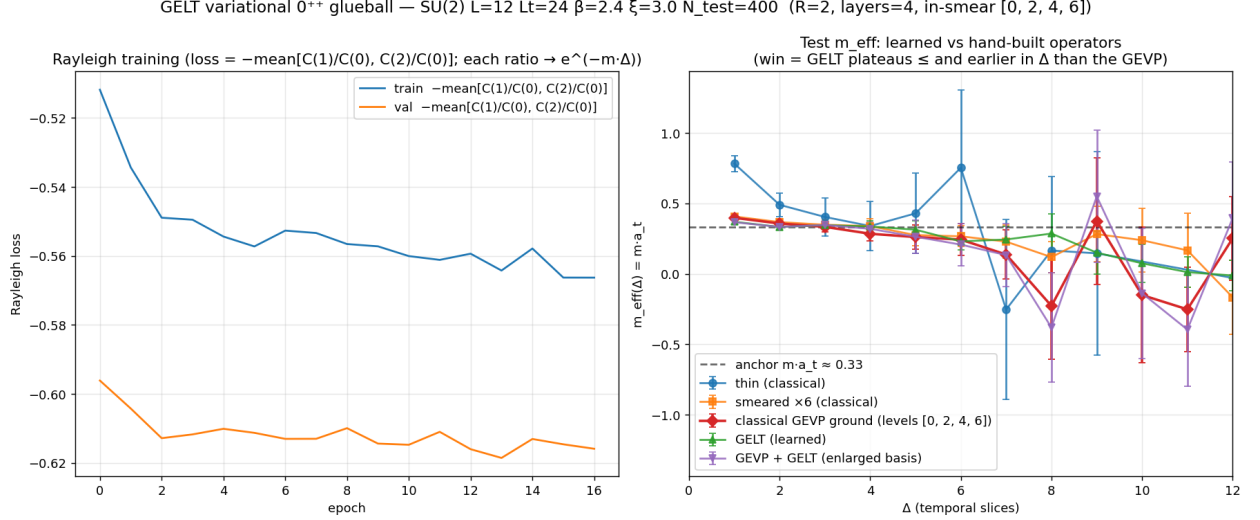


FIG. 5. The learned variational operator. Left: Rayleigh training curves; the validation loss sits on the variational floor, $-0.6185 \leftrightarrow m_{a_t} \approx 0.329$ (validation below training is an estimator artifact — the 6-configuration batch estimator is noise-biased toward zero while the 200-configuration validation estimate is cleaner). Right: test effective masses. The learned operator (green) is flat on the anchor (dashed) from $\Delta = 1$, below the still-descending classical GEVP (red) at small Δ ; the enlarged GEVP+GELT basis (purple) collapses onto pure GELT.

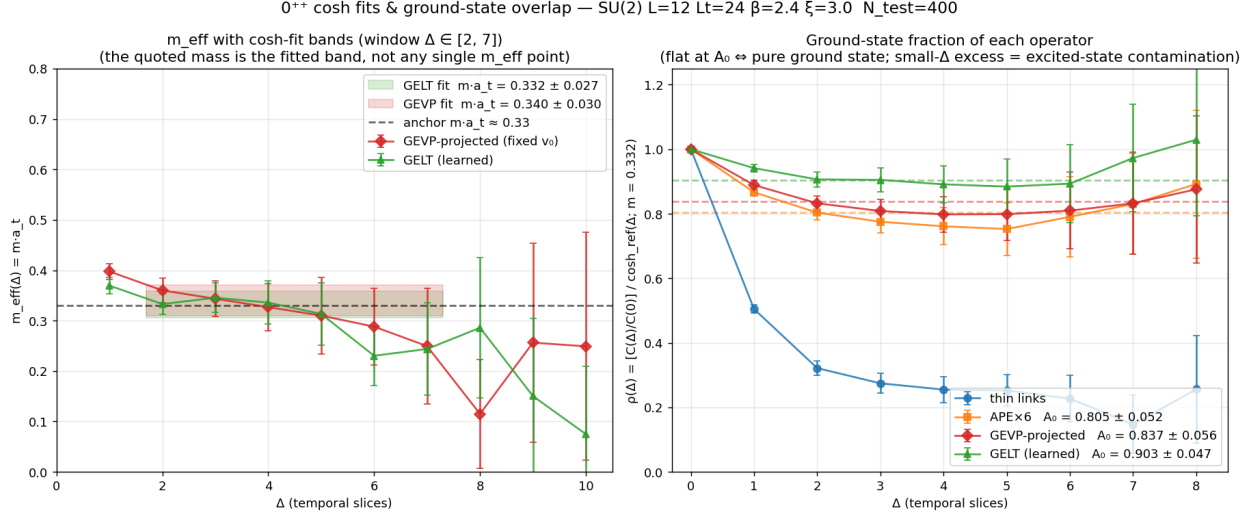


FIG. 6. The result in standard spectroscopic form. Left: test $m_{\text{eff}}(\Delta)$ for GELT (green) and the projected GEVP operator (red) with their fitted-mass bands over the shared window; the quoted mass is the band, not any single point, and both sit on the classical anchor (dashed). Right: the ground-state fraction $\rho(\Delta) = [C(\Delta)/C(0)] / \cosh_{\text{ref}}(\Delta)$, flat at A_0 for a pure interpolator; the excess at small Δ is excited-state contamination. Thin links collapse to $\rho \approx 0.25$ immediately, the classical operators descend through $\Delta \approx 3$ toward $A_0 \approx 0.80$ – 0.84 , and GELT is flat at $A_0 \approx 0.90$ from $\Delta = 1$.

$-\sum \alpha \log \alpha$. Since the trained operator is per timeslice (Sec. VC), $A(t, \vec{x})$ depends on slice t alone, its zero-momentum sum $\bar{A}(t)$ is a legitimate timeslice operator, and Eq. (8) applies verbatim:

$$\xi_A = 1/m_A \stackrel{?}{=} \xi(\beta) \quad (18)$$

with no ceiling and no geometric offset, because the correlator correlates a zero-mean *fluctuation* rather than a

moment of a kernel. Three consequences, in increasing order of interest: ξ_A should track $\xi(\beta)$; this measurement *does not exist for a convolution*, whose kernel is fixed so that $\delta A \equiv 0$ and $C_A \equiv 0$ identically; and the overlap A_0 of the attention field says whether *training* taught the routing to look at the physics or whether the equivariant construction alone is responsible.

That last distinction is essential, because the first consequence is close to a theorem — any local gauge-

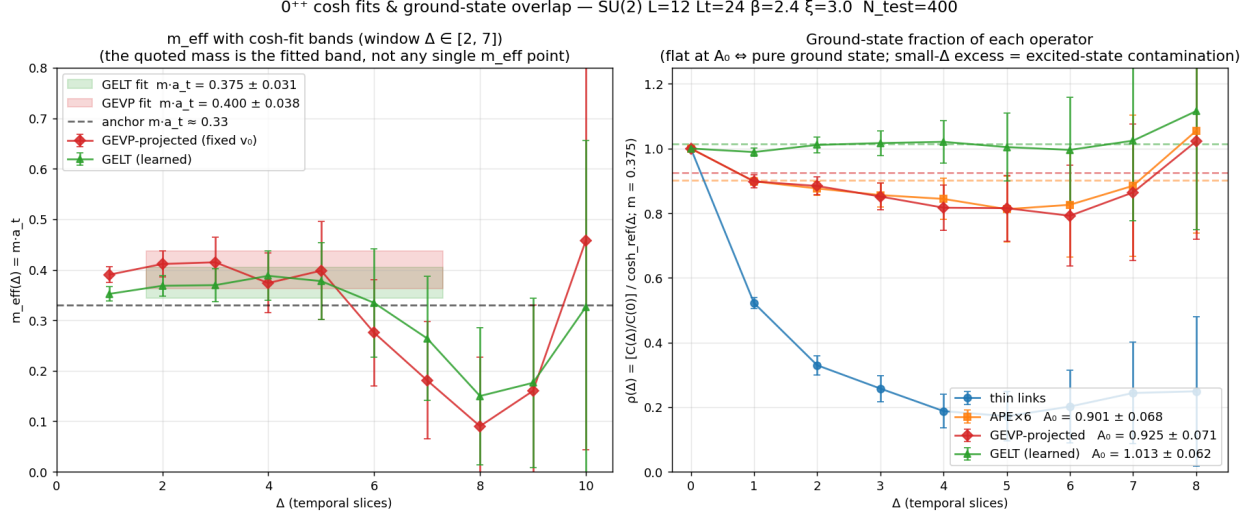


FIG. 7. Figure 6 repeated on the independently sampled replication ensemble. The overall mass scale sits $\sim 1\sigma$ higher — both operators agree on that shift, as they must — while the operator-quality ordering is unchanged: GELT flat at its A_0 from small Δ , the classical operators descending into it.

invariant scalar has a correlator decaying at the mass gap, a randomly initialized network’s attention field included. The design therefore has three arms: a trained network on its own ensemble; each trained network on *all* ensembles (so that operator quality is a property of the row and physics a property of the column); and a random-initialization control.

B. Testbed and protocol

The measurement needs a correlation length that is large compared with one lattice spacing and tunable over a real range. The anisotropic $SU(2)$ ensembles of Sec. V supply neither: a classical GEVP scan gives spatial correlation lengths $\xi_s = 1/(\xi m_{a,t})$ of 0.57, 0.83, 1.00, 1.10, 1.00 at $\beta = 2.1, 2.3, 2.4, 2.5, 2.7$ — never more than 1.1 spacings, so the whole object fits inside one grid cell. (The box is large, $L/\xi_s \approx 12$; it is the spacing that is coarse, because $\beta_s = \beta/\xi \approx 0.8$ is strong coupling.) Three-dimensional $\mathbb{Z}_2$ gauge theory [2] does supply both: it is dual to the 3D Ising model, has a second-order point at $\beta_c = 0.7614133$ (the image under the duality of the Ising $\beta_c^* = 0.221654626(5)$ [3]), and gives $\xi = 2.1 \rightarrow 6.4$ over four ensembles at $24^2 \times 48$, $N = 2000$. The price is that the conclusion is a statement about the method, not about $SU(N)$.

A fifth ensemble at $\beta = 0.7600$ was measured and is *excluded*. The dual determination of Sec. VII puts its true ξ at ≈ 10 against $L = 24$, so every operator there — classical, trained and random alike — is measuring the box and reads 50–58% low. It cannot discriminate between operators and is dropped, together with the network trained on it. Excluding it costs no dynamic range,

since $\beta = 0.7585$ already carries the largest correlation length the volume can hold. We note that the original finite-volume guard, $\xi \leq L/4$, admitted this point because it was evaluated against the *classical* $\xi \approx 4.7$ — a contaminated underestimate of the very quantity being bounded. A finite-volume cut has to be applied to an uncontaminated ξ , which is one more thing the dual supplies.

One variational operator per β was trained exactly as in Sec. V — per-timeslice (2D here), $R = 6$, four layers, two heads, four cumulative APE input levels (0, 4, 8, 16), Rayleigh loss — on `configs[:800]`; *all* measurements below use `configs[800:]`, so the evaluation set is unseen by every checkpoint, the diagonal included. Per (network, ensemble) pair and for each of the $4 \times 2 \times 3 = 24$ attention channels we stash α under `no_grad`, reduce it by Eq. (17), form $\hat{A}(t)$, project the channel basis onto its GEVP ground vector with a variational self-check (fall back to the best single channel whenever the projection interpolates worse than a basis member), fit Eq. (13) over $\Delta \in [2, 8]$, and take errors from a blocked jackknife with block 20. The classical smeared basis is re-measured on the *same* configurations so that differences are correlated. Two nulls are used: a config-scramble, in which each timeslice is drawn from an independently permuted configuration, which destroys both the temporal correlation and the per-configuration zero mode; and a time-shuffle, which is *not* a null but a zero-mode consistency check, since permuting timeslices preserves each configuration’s mean and leaves a flat pedestal $(2\xi - 1)/N_t$.

TABLE V. Correlation lengths and ground-state overlaps of the attention field on 1200 unseen configurations per ensemble, against the classical smeared basis measured on the same configurations and against a randomly initialized network of identical architecture. ΔA_0 is a blocked jackknife of the trained – random *difference* on shared configurations; $\delta A/A$ is the site-level fluctuation-to-mean ratio, median over the 24 channels.

β	ξ_{class}	ξ_A trained	ξ_A random	A_0 tr./rnd./cl.	ΔA_0
0.7450	2.04(14)	2.24(12)	2.08(13)	0.85/0.70/0.58	+0.144(17)
0.7520	2.34(16)	2.76(13)	2.46(16)	0.78/0.67/0.58	+0.110(20)
0.7560	4.08(27)	4.48(30)	4.08(26)	0.74/0.53/0.45	+0.204(12)
0.7585	5.59(34)	6.64(32)	5.47(31)	0.75/0.49/0.41	+0.267(11)

C. Results

Table V is the diagonal run on 1200 unseen configurations per ensemble.

$$\text{Pearson}(\xi_A, \xi_{\text{class}}) = \mathbf{0.9966}, \quad \text{slope } 1.21, \quad (19)$$

over a factor 2.74 of classical dynamic range at 5–7% precision on both sides. At $\beta = 0.7585$, the ensemble with the largest correlation length the volume can hold, the normalized correlator profiles read

$C(\Delta)/C(0)$	$\Delta = 1$	2	3	4
classical basis	0.441	0.303	0.230	0.192
trained attention	0.695	0.565	0.477	0.406
random attention	0.508	0.351	0.271	0.223
time-shuffle	0.236	0.229	0.231	0.234

— the trained attention field is a visibly cleaner exponential than either the classical basis or the untrained network. The last row is the zero-mode consistency check rather than a null: permuting timeslices destroys the temporal correlation but preserves each configuration’s mean, leaving the flat pedestal $(2\xi_A - 1)/N_t \approx 0.26$ that is observed. The actual null is the config-scramble, which is flat at the 10^{-2} level with no fittable mass (in the full cross matrix, 26 of 30 null cells do not fit at all and the remaining four return $|A_0| \leq 0.0066$).

No memorization. Table VI evaluates all four networks on all four ensembles. The matrix reads across, not down:

$$\frac{\text{spread within a row (evaluation)}}{\text{spread within a column (training } \beta)} = \frac{1.477}{0.123} = \mathbf{12.0}, \quad (20)$$

and the column means correlate with the classical ξ at Pearson 0.9992. The mean diagonal advantage in A_0 is only +0.038 and is inconsistent in sign, against a mean trained-minus-random of +0.156: the learning effect is 4× the β -matching effect, and the best network overall (**train@0.7585**, row mean $A_0 = 0.786$ against 0.590 for random) is best on three of four ensembles including ones

TABLE VI. ξ_A from the cross-evaluation matrix (600 unseen configurations per cell). Rows are the coupling at which the network was trained, columns the ensemble it is evaluated on. Entries are the multi-channel GEVP arm; at (trained 0.7560, evaluated 0.7520) the single-channel arm failed its fit-quality guard while the multi-channel projection resolved cleanly ($A_0 = 0.79$), and the latter is what this table quotes throughout.

trained at	evaluation ensemble β			
	0.7450	0.7520	0.7560	0.7585
0.7450	2.24	2.73	4.53	5.73
0.7520	2.33	2.77	4.64	5.74
0.7560	2.36	2.91	4.71	6.06
0.7585	2.31	2.80	4.68	6.50
random init	2.05	2.47	4.32	5.42
classical	2.05	2.36	4.21	5.48

it never saw. Every trained network beats the random arm on every ensemble. The networks read the configuration in front of them.

D. What is established, and what is not

The random-init arm *also* tracks ξ , at Pearson 0.9996 — if anything more tightly. That is the predicted outcome, and it forces a clean partition:

Structural claim (established). Attention maps in a gauge-equivariant network are bona fide lattice operators, with a mass, an overlap, and a spectral decomposition. This is established by $\xi_A \approx \xi$ and holds already at initialization; it is a property of the architecture. It is also the premise of the whole interpretability program made quantitative, and it has no analogue for a fixed kernel, for which $\delta A \equiv 0$ by construction.

Learning claim (established, by a different statistic). **Training makes those operators good: ΔA_0 from +0.11 to +0.27 (5.4 σ to 23.5 σ , correlated differences)**, the site-level fluctuation $\delta A/A$ raised by factors 14–46 over random initialization, and the transfer test above. *Not* by ξ_A . “The network discovers the correlation length” is not a supportable statement; **“training removes excited-state contamination from the attention field”** is.

The trained attention field is, by the A_0 measure, a better operator than the hand-built smeared basis at every β , and increasingly so toward the critical point ($0.85 \rightarrow 0.65$ for the attention against $0.58 \rightarrow 0.40$ for the classical basis).

On the overshoot. The trained arm reads ξ *above* the classical operator (slope 1.20) while the random arm agrees with it (slope 0.95), which inverts the naive expectation. It is not a contradiction, because the classical operator is not ground truth: it is one more contaminated operator, and by Eq. (8) excited states add faster-

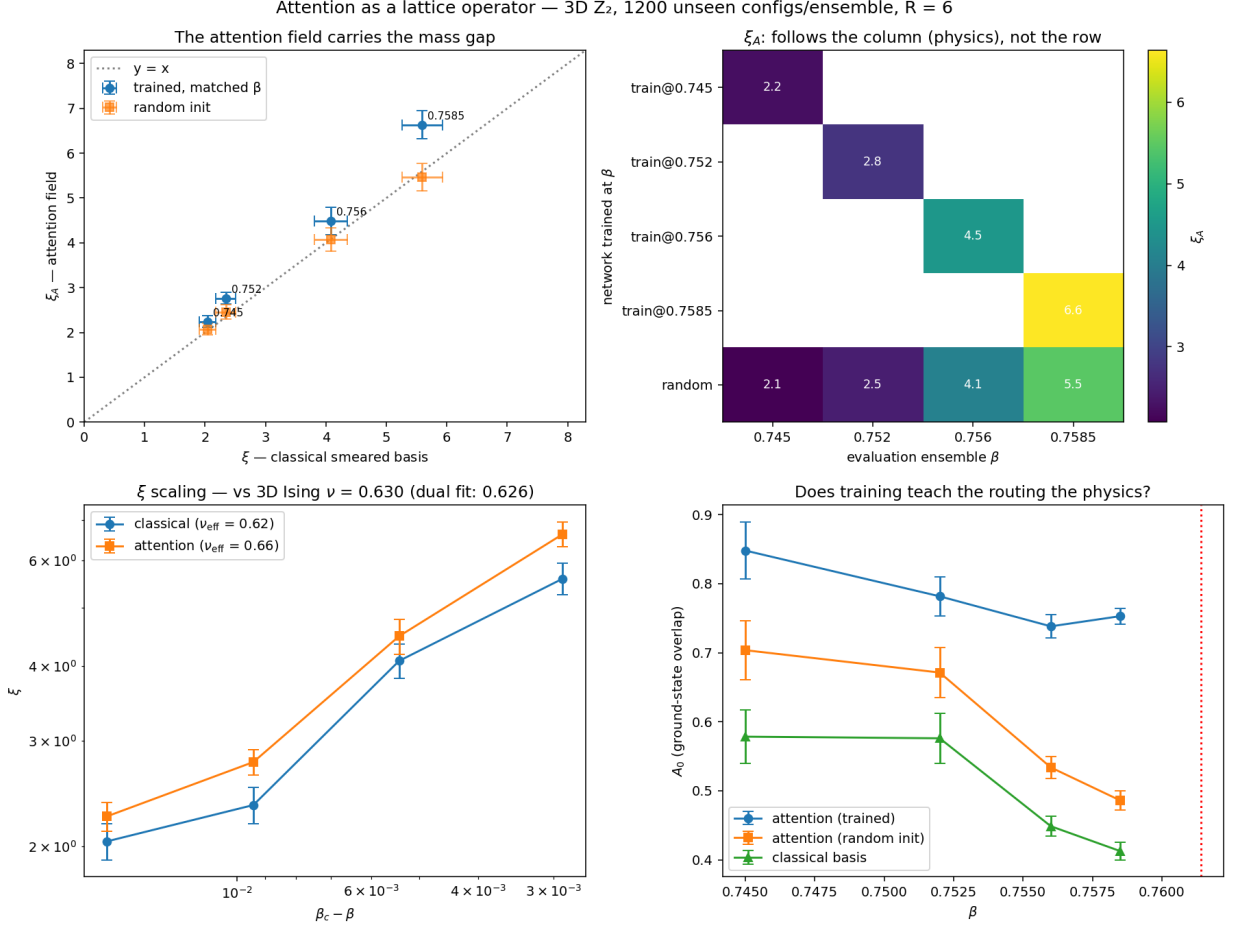


FIG. 8. The attention field as a lattice operator, 3D $\mathbb{Z}_2$, 1200 unseen configurations per ensemble. Top left: ξ_A against the classical ξ measured on the same configurations, for the matched trained network and for random initialization; dotted line is $y = x$. Top right: the cross-evaluation matrix — ξ_A follows the column (the physics), not the row (the training β). Bottom left: both lengths against $\beta_c - \beta$; the effective exponents, 0.62 classical and 0.66 attention, agree with the 0.60 returned by the dual ground truth fitted identically on the same four couplings (see text — this is a consistency check, not a measurement of ν). Bottom right: ground-state overlap A_0 of the trained attention, the random attention, and the classical basis.

decaying exponentials, so contamination biases ξ low. The operator with the highest A_0 should read the largest ξ , and it does. We therefore do not quote $|\xi_A - \xi_{\text{class}}|$ as an accuracy metric — agreement with a contaminated reference is not accuracy. Which of the two is closer to the true gap is not decidable from Table V, and Sec. VII decides it against an external standard.

Limitations. One theory, one lattice family, and the conclusion is about the method rather than about $SU(N)$. Since α is a softmax output, a network with larger score magnitudes has sharper and hence more variable attention regardless of what it learned, so $\delta A/A$ is descriptive and the load-bearing learning statistic is A_0 ; closing that confound properly requires an entropy-matched random control. The $R = 6$ operators are also of uneven quality across β , which the cross-evaluation design turns into a nuisance parameter — operator quality

is a property of the row, physics of the column — but which the diagonal alone would still inherit.

VII. GROUND TRUTH FROM THE DUAL ISING MODEL

Everything in Sec. VI compares one operator with another. The question it cannot answer — which of them is closer to the true mass gap — requires a reference that is not itself a variational operator. Three-dimensional $\mathbb{Z}_2$ gauge theory has one, exactly.

A. Why the dual is a reference and not another operator

Wegner’s duality [2] maps 3D $\mathbb{Z}_2$ gauge theory at β onto the 3D Ising model at

$$\beta^* = -\frac{1}{2} \ln \tanh \beta \quad \Longleftrightarrow \quad \sinh 2\beta \sinh 2\beta^* = 1, \quad (21)$$

plaquettes of Λ onto bonds of the dual lattice, and the confined phase onto the *broken* phase. Three consequences matter here.

First, the mass gap is the same number, not a proxy: the 0^{++} glueball is the lightest state of the $\mathbb{Z}_2$ -even sector, which is the lightest Ising excitation. Second — and this is what makes the reference uncontaminated in the sense Sec. VI needs — in the broken phase $\langle \sigma \rangle \neq 0$, so the Ising *order parameter* interpolates that state with near-unit overlap. On the gauge side σ is the ’t Hooft disorder operator: non-local in the links, and therefore not a member of any smeared-loop variational basis the classical measurement could have chosen. The reference is outside the family being judged, not a better member of it. Third, it is essentially free: an Ising sweep is two shift stencils, so statistics and volumes far beyond the reach of the gauge ensembles cost minutes.

We measure two dual operators, both analysed through the *same* code path as the gauge side (same connected correlator, same $\Delta \in [2, 8]$ cosh fit, same blocked jack-knife), so that nothing in the comparison differs except the operator: the wall magnetization σ , and the temporal-bond wall energy ε_t , which is the literal dual of the spatial plaquette the glueball operator is built from. Two volumes are used: the *matched* 48×24^2 , which is the dual of the gauge torus, and a *large* 96×48^2 .

B. Validating the duality before using it

Differentiating the duality relation between the two partition functions with respect to β gives a prediction with no free parameter,

$$\langle P \rangle(\beta) = \tanh \beta + \frac{1 - \langle s_i s_j \rangle(\beta^*)}{\sinh 2\beta}, \quad (22)$$

whose left side is measured on the production gauge ensembles and whose right side is measured in the Ising simulation. Nothing is fitted and the two sides are produced by unrelated code. Equation (22) is verified analytically — feeding the Ising low-temperature series $1 - \langle ss \rangle \approx 4t^6$ ($t = \tanh \beta$) into it returns the gauge strong-coupling series $\langle P \rangle \approx t + 2t^5$, whose coefficient counts the cubes containing a plaquette in 3D — and numerically, at -0.9σ and $+1.5\sigma$ on the two couplings whose Markov chains satisfy the sampling criterion below.

The sharper validation is that the dual reproduces 3D Ising criticality. Fixing the amplitude in $\xi = \xi_0 t^{*- \nu}$,

$t^* = (\beta^* - \beta_c^*)/\beta_c^*$, from the *single* lowest coupling and taking $\nu = 0.629971(4)$ from the conformal bootstrap [4], every other point becomes a prediction:

β	0.7450	0.7520	0.7560	0.7585
ξ predicted (fixes ξ_0)	3.015	4.285	6.341	
ξ measured	2.114(45)	2.962(50)	4.251(62)	6.021(78)
deviation	—	−1.8%	−0.8%	−5.1%

Anchoring on the coupling furthest from β_c minimizes finite-volume contamination but maximizes corrections to scaling, so the residual at the closest coupling should not be over-read; a two-parameter fit to the same four points returns $\nu = 0.60$, against the bootstrap 0.630. What the test establishes is that the dual is on the 3D Ising scaling curve to a few percent, which is the precision the comparison below needs, not that it measures ν .

Sampling, and why the matched volume is unavailable near β_c . In the broken phase the slowest mode is *not* the critical one: tunnelling between magnetization sectors has a barrier set by the interface area, so its auto-correlation time is exponential in the volume rather than ξ^z . At $\beta = 0.760$ on the matched volume, $\xi^2 \approx 76$ sweeps against a measured $\tau_{\text{int}}(M) \approx 1.6 \times 10^3$. Thermalization and decorrelation must therefore be fixed from τ_{int} of the magnetization, not of the energy and not from ξ^z ; we require $n_{\text{therm}} \geq 20 \tau_{\text{int}}(M)$ and $n_{\text{skip}} \geq 2 \tau_{\text{int}}(M)$. Because τ_{int} is returned per *measurement* and floors at $1/2$ for a decorrelated chain, it has to be determined once in a pilot at $n_{\text{skip}} = 1$, where it is directly in sweeps; re-measuring it inside a loop that adjusts n_{skip} makes the criterion self-referential and unsatisfiable.

This is also where the matched volume runs out. At $\beta \geq 0.756$ that box tunnels on 6% to 48% of consecutive measurements, and the consequence is not a sampling error to be averaged away: a box in which the broken phase is not cleanly broken admits the tunnelling mode into the σ correlator as a genuine light state, which inflates ξ — visibly, since the matched volume then reads *above* the large one (7.80 against 6.02 at $\beta = 0.7585$) where finite volume can only squeeze. More sweeps sample that mode better rather than removing it. We therefore take the reference from the large volume wherever σ tunnels, and note that ε_t , being $\mathbb{Z}_2$ -even, is blind to the sector and unaffected throughout. The tunnelling barrier grows with volume, so the *larger* lattice is the better-behaved one here, inverting the usual intuition.

C. Which operator is right

Table VII is the comparison. **The trained attention field is the only arm consistent with the exact answer;** the classical smeared basis and the untrained network are low by 9% and 8%, and the ordering is exactly the one their A_0 predicts. The single coupling that discriminates most

TABLE VII. Fractional deviation of each gauge-side operator from the dual ground truth, with the significance of the deviation in parentheses. The reference is the large-volume measurement: near β_c the matched box tunnels between magnetization sectors (below), so the σ correlator there is not measuring the σ particle. Since finite volume can only push the gauge values *down*, this choice cannot manufacture the trained arm’s agreement, though it can inflate the deficits of the other two. $\beta = 0.7600$ is excluded: its true $\xi \approx 10$ exceeds what $L = 24$ can hold, so all three operators are 50–58% low together and the point discriminates nothing.

β	ξ exact	classical	attention trained	attention random
0.7450	2.114(45)	−3.5% (0.5 σ)	+6.1% (1.0 σ)	−1.8% (0.3 σ)
0.7520	2.962(50)	−20.9% (3.6 σ)	−6.7% (1.4 σ)	−17.0% (3.1 σ)
0.7560	4.251(62)	−4.1% (0.6 σ)	+5.5% (0.8 σ)	−4.1% (0.7 σ)
0.7585	6.021(78)	−7.1% (1.2 σ)	+10.2% (1.9 σ)	−9.2% (1.7 σ)
<i>mean</i>		−8.9%	+3.8%	−8.0%

sharply is $\beta = 0.7520$, where the classical basis is 3.6σ low; at 0.7450 and 0.7560 the gauge error bars (± 0.13 to ± 0.30) are too wide to separate a 5% effect, so the statement rests on two of the four points.

This settles the overshoot of Sec. VI, and it settles it in the direction A_0 predicted: **the trained operator reads ξ high relative to the classical basis because the classical basis reads it low**, and the random arm’s apparent agreement with the classical operator was two comparably biased operators landing in the same place rather than two accurate ones. Against an external standard they are −10.2% and −11.1%.

The mechanism can be isolated on the dual side alone, with every confound removed — same lattice, same configurations, same fit window, one operator swapped:

β	0.7450	0.7520	0.7560	0.7585
$A_0(\sigma)$	0.95	0.95	0.96	0.97
$A_0(\varepsilon_t)$	0.61	0.49	0.40	0.36
$\xi(\varepsilon_t)/\xi(\sigma) - 1$	−10.4%	−13.6%	−19.1%	−31.5%

Contamination biases ξ low, monotonically in A_0 . This is the content of Eq. (8) demonstrated rather than argued, and it is what licenses reading ΔA_0 in Table V as a statement about accuracy.

Two corrections to what we would otherwise have reported. The dual also overturns two claims that the gauge-side data alone supported. First, an earlier version of this work treated the non-monotonic excursion $\xi(0.7585) > \xi(0.760)$ — seen in the classical measurement and in the attention, on the same configurations — as the strongest single piece of evidence, on the grounds that two quantities which turn around at the same place do not correlate for trivial reasons. The dual is monotonic at *both* volumes. The excursion was finite volume at $\beta = 0.760$, inherited by both operators precisely because

TABLE VIII. Attention statistics and head ablation of the trained $SU(2)$ glueball operator on a held-out $N = 32$ ensemble. The ℓ_{att} error is the *site-to-site* spread — the configuration-dependent part, which a fixed convolution kernel could not supply. Spearman errors are configuration level. Baseline Rayleigh loss −0.538; a positive Δloss means removing the head makes the operator worse.

layer/head	ℓ_{att}	entropy	Spearman(ℓ, o)	Δloss
1/0	1.69(15)	2.24	−0.139(1)	−0.0023(10)
1/1	1.87(6)	1.04	−0.009(1)	−0.0002(5)
2/0	1.93(5)	0.35	−0.086(1)	−0.0025(11)
2/1	2.000(0)	0.004	+0.397(2)	−0.0003(17)
3/0	2.00(1)	0.07	−0.088(1)	−0.0010(14)
3/1	0.30(18)	0.89	−0.328(2)	+0.0787(145)
4/0	1.59(21)	1.25	−0.230(2)	+0.0005(55)
4/1	1.44(12)	2.22	−0.564(2)	+0.0058(36)

both were measured on the same $L = 24$ configurations, and the argument is withdrawn. Second, the effective exponent from a scan including that point is 0.43 (classical) and 0.48 (attention), which we would have reported as evidence that $L = 24$ is not the asymptotic scaling regime. It is not: with the out-of-box point removed the same fit gives 0.622 and 0.656, against 0.60 from the dual fitted identically on the same four couplings. The dual arm returns 0.63 either way, being the only arm measured at a volume that can hold $\xi \approx 11$ — which is what distinguishes “outside the scaling window”, where the dual would have moved too, from “one point outside the box”, where it does not.

VIII. WHAT THE LEARNED OPERATOR ATTENDS TO

The validation above establishes *that* the learned glueball operator is better; the attention readout asks *how*. Loading the trained $SU(2)$ operator unchanged and running it in inference over a fresh $N = 32$ ensemble sampled with a new seed (768 timeslices, 1728 sites each), we accumulate per (layer, head) the attention range $\ell_{\text{att}}(x) = \sum_n \alpha_n |\Delta x_n|_1$, the offset entropy, a per-site axial anisotropy, the Spearman correlation of $\ell_{\text{att}}(x)$ against the smeared spatial-plaquette density $o(x)$, and a head-ablation intervention in which the head’s column of the output mixing matrix is zeroed and the degradation of the Rayleigh loss is measured as a correlated delete-one jackknife. On this previously unseen chain the operator returns a Rayleigh loss of −0.538 — a third independent realization of the ensemble-specific floor — confirming that the weights are not tuned to their training chain. Table VIII and Fig. 9 summarize.

Three findings. *One head does the work*: layer 3, head 1 costs $+0.0787 \pm 0.0145$ in Rayleigh loss when removed

— 5.4σ , an order of magnitude more than any other head and roughly a seventh of the entire signal — with nothing else above 2σ in the positive direction. That same head is the only *local* one in the network, $\ell_{\text{att}} = 0.30$ with 71% of its mass on the site itself, against six heads at or near the maximal range. The optimal operator is not a democratic ensemble of scales but one content-dependent local head embedded in a stack of long-range ones.

Attention reaches further where the action is dense: the highest-entropy head, i.e. the most genuinely distributed one, has $\ell_{\text{att}}(x)$ anticorrelating with the smeared plaquette density at Spearman -0.56 . Since large $o(x)$ means *small* action density, the head extends its reach on disordered, high-action regions and contracts on smooth vacuum — measured against a physical field rather than asserted from a picture.

The learned operator is only approximately a cubic scalar: four heads sit at axial anisotropy $\gtrsim 1.36$ of a maximum 1.41, and their selected axis is globally frozen — three heads select the same spatial axis at 100.0% of the 1.3 million site-slice samples. Two facts keep this from undermining Sec. V: the axis-locked heads are exactly the low-entropy, ablation-irrelevant ones, while the two heads that matter physically are the two most isotropic; and a spin-2 admixture contaminates the correlator with *heavier* states, biasing an extracted mass upward, so it cannot manufacture the overlap excess of Eq. (16). What it does identify is a concrete and untested improvement: an explicit A_1^{++} projection, averaging the operator over the 24 lattice rotations.

IX. DISCUSSION

Two threads run through these results, and they meet.

The first is that *a gauge-equivariant network is a legitimate lattice operator, so the variational machinery of spectroscopy applies to it unchanged*. The Rayleigh quotient is a physics objective, not a surrogate loss; its converged value is a mass; the transfer-matrix bound holds at every step of training precisely because equivariance is exact rather than learned. Under that discipline — and under the single-timeslice restriction the same theory demands — one small learned operator ($\sim 1.6 \times 10^4$ real parameters) carries more ground-state weight than the optimal linear combination of a four-level Morningstar-Pearson basis built from the same smearing levels, $\Delta A_0 = +0.078 \pm 0.022$ over two independently sampled ensembles, at a consistent mass. The comparison is deliberately conservative: the network is given exactly the classical basis's inputs, judged by the classical figure of merit, on configurations no checkpoint has seen, with the difference jackknifed rather than the two errors added.

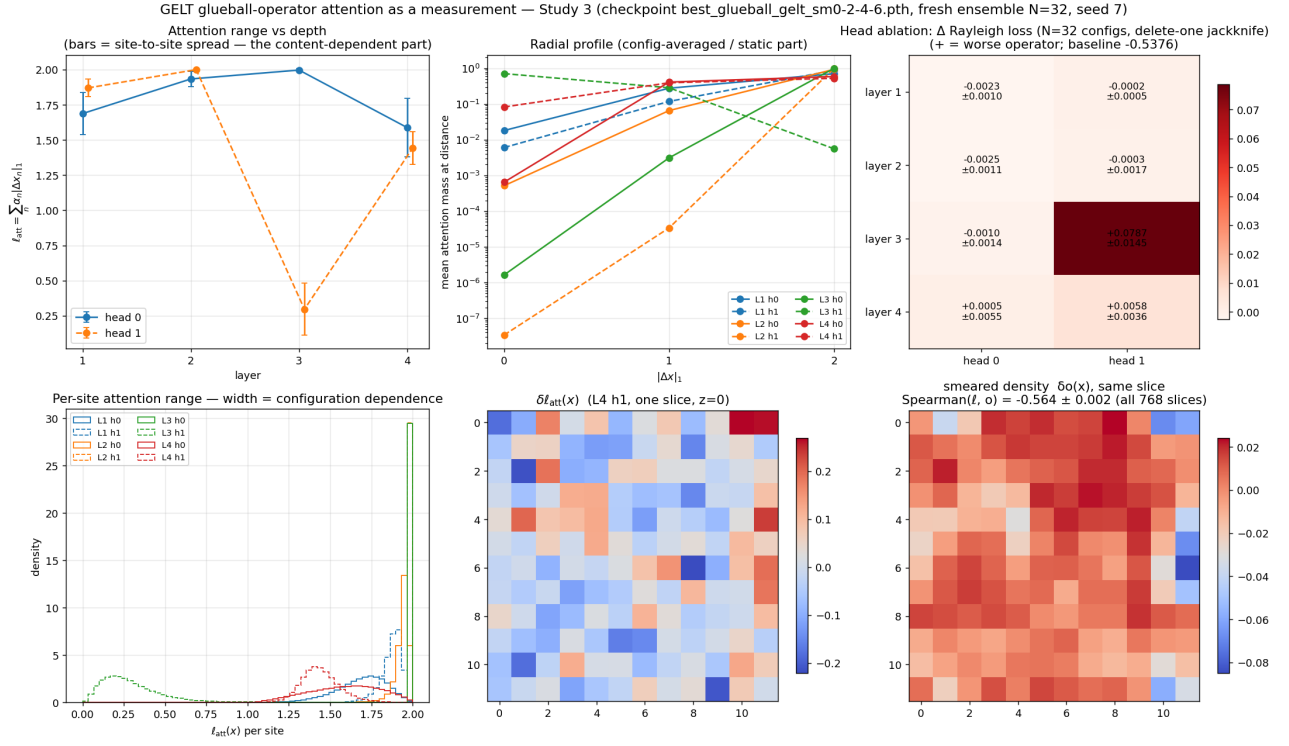
The second is that *the attention map is itself an observable*. This is not an interpretability heuristic. Gauge

invariance of the score makes any reduction of an attention row a local scalar field; single-timeslice locality makes its correlator spectral; so the attention field has a mass, an overlap, and a spectral decomposition, and the measurement has no analogue for a convolution, whose kernel does not fluctuate at all. In 3D $\mathbb{Z}_2$ this yields ξ_A tracking ξ at Pearson 0.9966, and a control matrix in which ξ_A is $12\times$ more sensitive to the ensemble in front of the network than to the coupling it was trained at. We are explicit that the structural half of this holds at random initialization; what training buys is measured separately, and it is ground-state purity.

The third thread is that in this theory the two can be checked against something external. Because 3D $\mathbb{Z}_2$ is exactly dual to the 3D Ising model, the same mass gap is measurable with the dual order parameter — an operator that is non-local in the gauge variables and so could never have been a member of the basis being compared — and the trained attention field turns out to be the only arm consistent with it while the classical smeared basis and the untrained network are 9% and 8% low. That converts ΔA_0 from a statement about operator quality into a statement about accuracy, and it is the sense in which the learned operator is not merely different from the standard one but better. The same reference cost us two claims we would otherwise have made (Sec. VII), which is the point of having one.

The two threads meet in the head analysis: the operator that wins on overlap turns out to concentrate its advantage in a single local, content-dependent head, while several of its long-range heads have collapsed to fixed kernels and to frozen spatial axes. That is an actionable diagnosis rather than a visualization — it says that at $R = 2$ the range hierarchy has nowhere to go, that an A_1^{++} rotation average is the cheapest available improvement, and that the ablation-irrelevant heads are capacity to reallocate.

Three directions follow. *Symmetry:* projecting the operator onto the cubic A_1^{++} representation, which the readout shows is currently only approximate. *Scale:* pushing the $\mathbb{Z}_2$ measurement to $L \geq 48$, which the dual shows is what it would take to add a point at $\xi \approx 10$ — not to settle the excursion, which the dual has already settled, but to extend the lever arm — and repeating the attention correlator on the $SU(2)$ operator, where the machinery already exists and only the correlation length does not. *Ground truth:* $SU(N)$ has no dual, so the calibration of Sec. VII is available in $\mathbb{Z}_2$ and nowhere else; what transfers is not the number but the lesson that A_0 is the quantity to trust when two operators disagree about a mass. *Scope:* $SU(3)$ and excited states, where the same Rayleigh construction generalizes to a multi-operator GEVP over several learned channels, and where the cost of the transport tensor — the binding constraint of the architecture — makes offset-chunked attention the next piece of engineering.



All computations were performed with the repository's own implementations of the lattice primitives, samplers, spectroscopy and network layers; the unit-test suite covering gauge equivariance, transport covariance, sampler exactness and correlator arithmetic passes in full (111 tests). The dual-model ground truth of Sec. VII is checked by exact enumeration of the 2^{16} states of a 4×4 Ising lattice and by the parameter-free relation (22) against a gauge simulation run by separate code.

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